Q.No:6

DS	Session	Linked List	Question Information	Level 1 Challenge 36

Question description

A long time ago, there was a desolate village in India. The ancient buildings, streets, and businesses were deserted. The windows were open, and the stairwell was in disarray. You can be sure that it will be a fantastic area for mice to romp about in! People in the community have now chosen to provide high-quality education to young people in order to further the village's growth.

As a result, they established a programming language coaching centre. People from the coaching centre are presently performing a test. Create a programme for the GetNth() function, which accepts a linked list and an integer index and returns the data value contained in the node at that index position.

Example

Input: 1->10->30->14, index = 2
Output: 30
The node at index 2 is 30

Constraints

1< N < 1000

1< X < 1000

1 < I < 1000

Input Format
First line contains the number of datas- N.
Second line contains N integers(the given linked list).
Third Line Index I

Output Format
First Line indicates the linked list

second line indicates the node at indexing position

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
4
1 2 3 4
2
```

EXPECTED OUTPUT

```
Linked list:-->4-->3-->2-->1
Node at index=2:3
```

Test Case 2

INPUT (STDIN)

```
4
11 21 31 41
3
```

EXPECTED OUTPUT

```
Linked list:-->41-->31-->21-->11
Node at index=3:21
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
struct node
```

Test Case 2

KEYWORD

```
struct node *next;
```

Test Case 3

KEYWORD

```
int GetNth(struct node*
head,int index)
```

Code:

```
#include <stdio.h>
#include <stdlib.h>

struct node
{
int data;
struct node *next;
};

struct node* insertAtBeginning(struct node *head, int data)
{
struct node *newNode;

newNode = (struct node *)malloc(sizeof(struct node));
```

```
newNode->data = data;
newNode->next = head;

return newNode;
}

void display(struct node *head)
{
struct node *temp = head;

printf("Linked list:-->");

while (temp != NULL)
{
printf("%d", temp->data);

if (temp->next != NULL)
printf("-->");

temp = temp->next;
}

printf("\n");
}

int GetNth(struct node *head, int index, int n)
{
struct node *temp = head;

int target = n - 1 - index;
int count = 0;

while (temp != NULL)
{
if (count == target)
return temp->data;

count++;
temp = temp->next;
}

return -1;
}

int main()
{
int n, data, index;
struct node *head = NULL;

printf("Input value:\n");
```

```c
scanf("%d", &n);

for (int i = 0; i < n; i++)
{
scanf("%d", &data);
head = insertAtBeginning(head, data);
}

scanf("%d", &index);

display(head);

printf("Node at index=%d:%d\n",
index, GetNth(head, index, n));

return 0;
}
```

Output:

Test _case 1:

```
Input value:
4
1 2 3 4
2
Linked list:-->4-->3-->2-->1
Node at index=2:3
```

Test _case 2:

```
aadityajaiswal@Aadityas-MacBook-Air linked list % ~/Users/
Input value:
4
11 21 31 41
3
Linked list:-->41-->31-->21-->11
Node at index=3:41
```